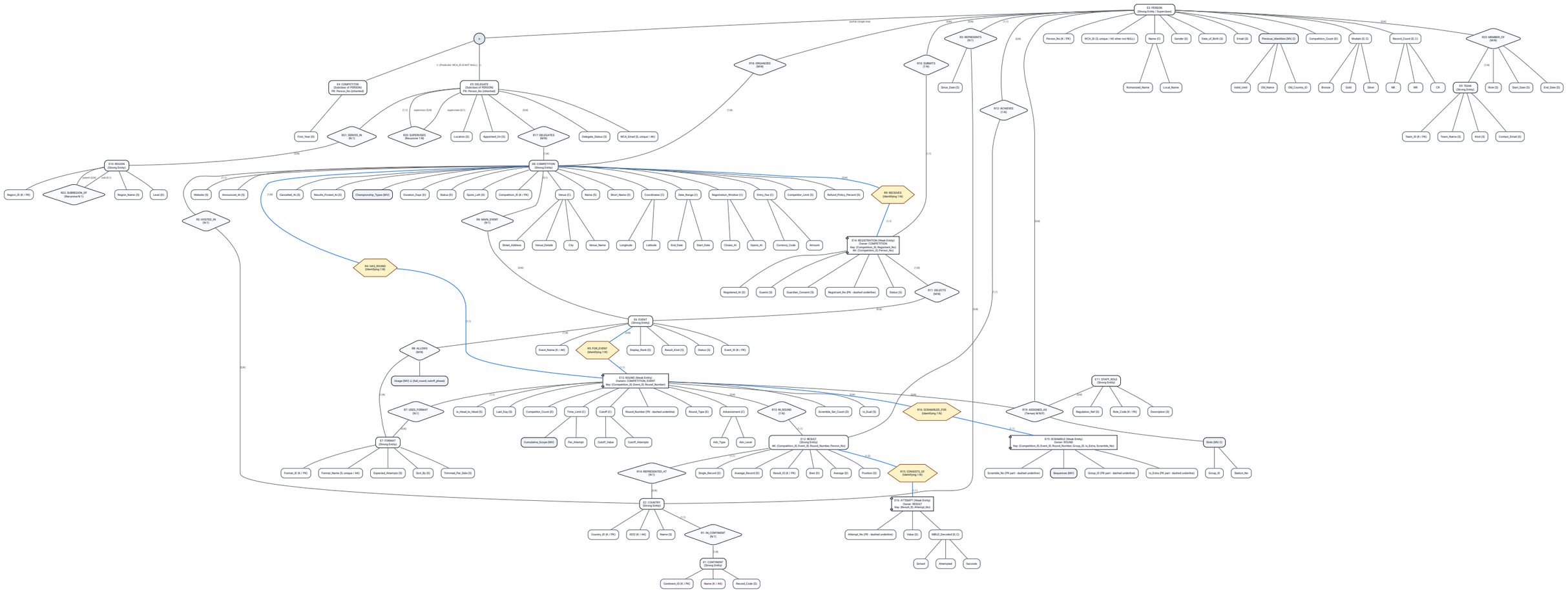
Option C: Complete Executable Mermaid EER Diagram (Entities E1–E16 & Relationships R1–R23)



Option C: Executable Mermaid Source Code (EER Shape Mapping)

```
flowchart TB
  classDef strong fill:#ffffff,stroke:#0f172a,stroke-width:2px,color:#0f172a,font-weight:bold
  classDef weak fill:#ffffff,stroke:#0f172a,stroke-width:4px,stroke-dasharray:0,color:#0f172a,font-weight:bold
  classDef rel fill:#f8fafc,stroke:#1e293b,stroke-width:2px,color:#0f172a,font-weight:bold
  classDef idrel fill:#fef3c7,stroke:#92400e,stroke-width:3.5px,color:#0f172a,font-weight:bold
  classDef spec fill:#e2e8f0,stroke:#0f172a,stroke-width:2px,color:#0f172a,font-weight:bold
  classDef attrS fill:#ffffff,stroke:#475569,stroke-width:1.2px,color:#0f172a
  classDef attrD fill:#ffffff,stroke:#475569,stroke-width:1.2px,stroke-dasharray:5 3,color:#0f172a
  classDef attrMV fill:#f1f5f9,stroke:#0f172a,stroke-width:3px,color:#0f172a

  %% ENTITIES E1-E16
  E1["E1: CONTINENT<br/>(Strong Entity)"]:::strong
  E2["E2: COUNTRY<br/>(Strong Entity)"]:::strong
  E3["E3: PERSON<br/>(Strong Entity / Superclass)"]:::strong
  E4["E4: COMPETITOR<br/>(Subclass of PERSON)<br/>PK: Person_No (inherited)"]:::strong
  E5["E5: DELEGATE<br/>(Subclass of PERSON)<br/>PK: Person_No (inherited)"]:::strong
  E6["E6: EVENT<br/>(Strong Entity)"]:::strong
  E7["E7: FORMAT<br/>(Strong Entity)"]:::strong
  E8["E8: COMPETITION<br/>(Strong Entity)"]:::strong
  E9["E9: TEAM<br/>(Strong Entity)"]:::strong
  E10["E10: REGION<br/>(Strong Entity)"]:::strong
  E11["E11: STAFF_ROLE<br/>(Strong Entity)"]:::strong
  E12["E12: RESULT<br/>(Strong Entity)<br/>AK: (Competition_ID, Event_ID, Round_Number, Person_No)"]:::strong

  E13[["E13: ROUND (Weak Entity)<br/>Owners: COMPETITION, EVENT<br/>Key: (Competition_ID, Event_ID, Round_Number)"]]:::weak
  E14[["E14: REGISTRATION (Weak Entity)<br/>Owner: COMPETITION<br/>Key: (Competition_ID, Registrant_No)<br/>AK: (Competition_ID, Person_No)"]]:::weak
  E15[["E15: SCRAMBLE (Weak Entity)<br/>Owner: ROUND<br/>Key: (Competition_ID, Event_ID, Round_Number, Group_ID, Is_Extra, Scramble_No)"]]:::weak
  E16[["E16: ATTEMPT (Weak Entity)<br/>Owner: RESULT<br/>Key: (Result_ID, Attempt_No)"]]:::weak

  %% SPECIALIZATION (Overlapping 'o', Partial single line)
  SPEC(("o")):::spec
  E3 ---|"partial (single line)"| SPEC
  SPEC ---|"⊂ (Predicate: WCA_ID IS NOT NULL)"| E4
  SPEC ---|"⊂"| E5

  %% E1 ATTRIBUTES
  E1 --- E1_A1(["<u>Continent_ID</u> (K / PK)"]):::attrS
  E1 --- E1_A2(["<u>Name</u> (K / AK)"]):::attrS
  E1 --- E1_A3(["Record_Code (S)"]):::attrS

  %% E2 ATTRIBUTES
  E2 --- E2_A1(["<u>Country_ID</u> (K / PK)"]):::attrS
  E2 --- E2_A2(["<u>ISO2</u> (K / AK)"]):::attrS
  E2 --- E2_A3(["Name (S)"]):::attrS

  %% E3 ATTRIBUTES
  E3 --- E3_A1(["<u>Person_No</u> (K / PK)"]):::attrS
  E3 --- E3_A2(["WCA_ID (S, unique / AK when not NULL)"]):::attrS
  E3 --- E3_A3(["Name (C)"]):::attrS
  E3_A3 --- E3_A3a(["Romanized_Name"]):::attrS
  E3_A3 --- E3_A3b(["Local_Name"]):::attrS
  E3 --- E3_A4(["Gender (S)"]):::attrS
  E3 --- E3_A5(["Date_of_Birth (S)"]):::attrS
  E3 --- E3_A6(["Email (S)"]):::attrS
  E3 --- E3_A7(["Previous_Identities (MV, C)"]):::attrMV
  E3_A7 --- E3_A7a(["Old_Name"]):::attrS
  E3_A7 --- E3_A7b(["Old_Country_ID"]):::attrS
  E3_A7 --- E3_A7c(["Valid_Until"]):::attrS
  E3 --- E3_A8(["Competition_Count (D)"]):::attrD
  E3 --- E3_A9(["Medals (D, C)"]):::attrD
  E3_A9 --- E3_A9a(["Gold"]):::attrD
  E3_A9 --- E3_A9b(["Silver"]):::attrD
  E3_A9 --- E3_A9c(["Bronze"]):::attrD
  E3 --- E3_A10(["Record_Count (D, C)"]):::attrD
  E3_A10 --- E3_A10a(["WR"]):::attrD
  E3_A10 --- E3_A10b(["CR"]):::attrD
  E3_A10 --- E3_A10c(["NR"]):::attrD

  %% E4 ATTRIBUTES
  E4 --- E4_A1(["First_Year (D)"]):::attrD

  %% E5 ATTRIBUTES
  E5 --- E5_A1(["Delegate_Status (S)"]):::attrS
  E5 --- E5_A2(["WCA_Email (S, unique / AK)"]):::attrS
  E5 --- E5_A3(["Location (S)"]):::attrS
  E5 --- E5_A4(["Appointed_On (S)"]):::attrS
```

```
%% E6 ATTRIBUTES
E6 --- E6_A1(["<u>Event_ID</u> (K / PK)"]):::attrS
E6 --- E6_A2(["<u>Event_Name</u> (K / AK)"]):::attrS
E6 --- E6_A3(["Display_Rank (S)"]):::attrS
E6 --- E6_A4(["Result_Kind (S)"]):::attrS
E6 --- E6_A5(["Status (S)"]):::attrS

%% E7 ATTRIBUTES
E7 --- E7_A1(["<u>Format_ID</u> (K / PK)"]):::attrS
E7 --- E7_A2(["Format_Name (S, unique / AK)"]):::attrS
E7 --- E7_A3(["Expected_Attempts (S)"]):::attrS
E7 --- E7_A4(["Sort_By (S)"]):::attrS
E7 --- E7_A5(["Trimmed_Per_Side (S)"]):::attrS

%% E8 ATTRIBUTES
E8 --- E8_A1(["<u>Competition_ID</u> (K / PK)"]):::attrS
E8 --- E8_A2(["Name (S)"]):::attrS
E8 --- E8_A3(["Short_Name (S)"]):::attrS
E8 --- E8_A4(["Venue (C)"]):::attrS
E8_A4 --- E8_A4a(["Venue_Name"]):::attrS
E8_A4 --- E8_A4b(["Street_Address"]):::attrS
E8_A4 --- E8_A4c(["Venue_Details"]):::attrS
E8_A4 --- E8_A4d(["City"]):::attrS
E8 --- E8_A5(["Coordinates (C)"]):::attrS
E8_A5 --- E8_A5a(["Latitude"]):::attrS
E8_A5 --- E8_A5b(["Longitude"]):::attrS
E8 --- E8_A6(["Date_Range (C)"]):::attrS
E8_A6 --- E8_A6a(["Start_Date"]):::attrS
E8_A6 --- E8_A6b(["End_Date"]):::attrS
E8 --- E8_A7(["Registration_Window (C)"]):::attrS
E8_A7 --- E8_A7a(["Opens_At"]):::attrS
E8_A7 --- E8_A7b(["Closes_At"]):::attrS
E8 --- E8_A8(["Entry_Fee (C)"]):::attrS
E8_A8 --- E8_A8a(["Amount"]):::attrS
E8_A8 --- E8_A8b(["Currency_Code"]):::attrS
E8 --- E8_A9(["Competitor_Limit (S)"]):::attrS
E8 --- E8_A10(["Refund_Policy_Percent (S)"]):::attrS
E8 --- E8_A11(["Website (S)"]):::attrS
E8 --- E8_A12(["Announced_At (S)"]):::attrS
E8 --- E8_A13(["Cancelled_At (S)"]):::attrS
E8 --- E8_A14(["Results_Posted_At (S)"]):::attrS
E8 --- E8_A15(["Championship_Types (MV)"]):::attrMV
E8 --- E8_A16(["Duration_Days (D)"]):::attrD
E8 --- E8_A17(["Status (D)"]):::attrD
E8 --- E8_A18(["Spots_Left (D)"]):::attrD

%% E9 ATTRIBUTES
E9 --- E9_A1(["<u>Team_ID</u> (K / PK)"]):::attrS
E9 --- E9_A2(["Team_Name (S)"]):::attrS
E9 --- E9_A3(["Kind (S)"]):::attrS
E9 --- E9_A4(["Contact_Email (S)"]):::attrS

%% E10 ATTRIBUTES
E10 --- E10_A1(["<u>Region_ID</u> (K / PK)"]):::attrS
E10 --- E10_A2(["Region_Name (S)"]):::attrS
E10 --- E10_A3(["Level (D)"]):::attrD

%% E11 ATTRIBUTES
E11 --- E11_A1(["<u>Role_Code</u> (K / PK)"]):::attrS
E11 --- E11_A2(["Description (S)"]):::attrS
E11 --- E11_A3(["Regulation_Ref (S)"]):::attrS

%% E12 ATTRIBUTES
E12 --- E12_A1(["<u>Result_ID</u> (K / PK)"]):::attrS
E12 --- E12_A2(["Best (D)"]):::attrD
E12 --- E12_A3(["Average (D)"]):::attrD
E12 --- E12_A4(["Position (D)"]):::attrD
E12 --- E12_A5(["Single_Record (D)"]):::attrD
E12 --- E12_A6(["Average_Record (D)"]):::attrD

%% E13 ATTRIBUTES
E13 --- E13_A1(["Round_Number (PK - dashed underline)"]):::attrS
E13 --- E13_A2(["Round_Type (D)"]):::attrD
E13 --- E13_A3(["Time_Limit (C)"]):::attrS
E13_A3 --- E13_A3a(["Per_Attempt"]):::attrS
E13_A3 --- E13_A3b(["Cumulative_Scope (MV)"]):::attrMV
E13 --- E13_A4(["Cutoff (C)"]):::attrS
E13_A4 --- E13_A4a(["Cutoff_Attempts"]):::attrS
E13_A4 --- E13_A4b(["Cutoff_Value"]):::attrS
```

```
E13 --- E13_A5(["Advancement (C)"]):::attrS
E13_A5 --- E13_A5a(["Adv_Type"]):::attrS
E13_A5 --- E13_A5b(["Adv_Level"]):::attrS
E13 --- E13_A6(["Scramble_Set_Count (D)"]):::attrD
E13 --- E13_A7(["Is_Dual (S)"]):::attrS
E13 --- E13_A8(["Is_Head_to_Head (S)"]):::attrS
E13 --- E13_A9(["Last_Day (S)"]):::attrS
E13 --- E13_A10(["Competitor_Count (D)"]):::attrD

%% E14 ATTRIBUTES
E14 --- E14_A1(["Registrant_No (PK - dashed underline)"]):::attrS
E14 --- E14_A2(["Status (S)"]):::attrS
E14 --- E14_A3(["Registered_At (S)"]):::attrS
E14 --- E14_A4(["Guests (S)"]):::attrS
E14 --- E14_A5(["Guardian_Consent (S)"]):::attrS

%% E15 ATTRIBUTES
E15 --- E15_A1(["Group_ID (PK part - dashed underline)"]):::attrS
E15 --- E15_A2(["Is_Extra (PK part - dashed underline)"]):::attrS
E15 --- E15_A3(["Scramble_No (PK part - dashed underline)"]):::attrS
E15 --- E15_A4(["Sequences (MV)"]):::attrMV

%% E16 ATTRIBUTES
E16 --- E16_A1(["Attempt_No (PK - dashed underline)"]):::attrS
E16 --- E16_A2(["Value (S)"]):::attrS
E16 --- E16_A3(["MBLD_Decoded (D, C)"]):::attrD
E16_A3 --- E16_A3a(["Solved"]):::attrD
E16_A3 --- E16_A3b(["Attempted"]):::attrD
E16_A3 --- E16_A3c(["Seconds"]):::attrD

%% RELATIONSHIPS R1-R23
R1{"R1: IN_CONTINENT<br/>(N:1)"}:::rel
E2 ---|"(1,1)"| R1
R1 ---|"(1,N)"| E1

R2{"R2: REPRESENTS<br/>(N:1)"}:::rel
E3 ---|"(1,1)"| R2
R2 ---|"(0,N)"| E2
R2 --- R2_A1(["Since_Date (S)"]):::attrS

R3{"R3: HOSTED_IN<br/>(N:1)"}:::rel
E8 ---|"(1,1)"| R3
R3 ---|"(0,N)"| E2

R4{{"R4: HAS_ROUND<br/>(Identifying 1:N)"}}:::idrel
E8 ===|"(1,N)"| R4
R4 ===|"(1,1)"| E13

R5{{"R5: FOR_EVENT<br/>(Identifying 1:N)"}}:::idrel
E6 ===|"(0,N)"| R5
R5 ===|"(1,1)"| E13

R6{"R6: MAIN_EVENT<br/>(N:1)"}:::rel
E8 ---|"(0,1)"| R6
R6 ---|"(0,N)"| E6

R7{"R7: USES_FORMAT<br/>(N:1)"}:::rel
E13 ---|"(1,1)"| R7
R7 ---|"(0,N)"| E7

R8{"R8: ALLOWS<br/>(M:N)"}:::rel
E6 ---|"(1,N)"| R8
R8 ---|"(1,N)"| E7
R8 --- R8_A1(["Usage (MV) ⊆ {full_round, cutoff_phase}"]):::attrMV

R9{{"R9: RECEIVES<br/>(Identifying 1:N)"}}:::idrel
E8 ===|"(0,N)"| R9
R9 ===|"(1,1)"| E14

R10{"R10: SUBMITS<br/>(1:N)"}:::rel
E3 ---|"(0,N)"| R10
R10 ---|"(1,1)"| E14

R11{"R11: SELECTS<br/>(M:N)"}:::rel
E14 ---|"(1,N)"| R11
R11 ---|"(0,N)"| E6

R12{"R12: ACHIEVES<br/>(1:N)"}:::rel
E3 ---|"(0,N)"| R12
R12 ---|"(1,1)"| E12
```

```
R13{"R13: IN_ROUND<br/>(1:N)"}:::rel
E13 ---|"(0,N)"| R13
R13 ---|"(1,1)"| E12

R14{"R14: REPRESENTED_AT<br/>(N:1)"}:::rel
E12 ---|"(1,1)"| R14
R14 ---|"(0,N)"| E2

R15{{"R15: CONSISTS_OF<br/>(Identifying 1:N)"}}:::idrel
E12 ===|"(1,5)"| R15
R15 ===|"(1,1)"| E16

R16{{"R16: SCRAMBLED_FOR<br/>(Identifying 1:N)"}}:::idrel
E13 ===|"(0,N)"| R16
R16 ===|"(1,1)"| E15

R17{"R17: DELEGATES<br/>(M:N)"}:::rel
E5 ---|"(0,N)"| R17
R17 ---|"(1,N)"| E8

R18{"R18: ORGANIZES<br/>(M:N)"}:::rel
E3 ---|"(0,N)"| R18
R18 ---|"(1,N)"| E8

R19{"R19: ASSIGNED_AS<br/>(Ternary M:N:P)"}:::rel
E3 ---|"(0,N)"| R19
E13 ---|"(0,N)"| R19
E11 ---|"(0,N)"| R19
R19 --- R19_A1(["Slots (MV, C)"]):::attrMV
R19_A1 --- R19_A1a(["Group_ID"]):::attrS
R19_A1 --- R19_A1b(["Station_No"]):::attrS

R20{"R20: SUPERVISES<br/>(Recursive 1:N)"}:::rel
E5 ---|"supervisor (0,N)"| R20
R20 ---|"supervisee (0,1)"| E5

R21{"R21: SERVES_IN<br/>(N:1)"}:::rel
E5 ---|"(1,1)"| R21
R21 ---|"(0,N)"| E10

R22{"R22: SUBREGION_OF<br/>(Recursive N:1)"}:::rel
E10 ---|"sub (0,1)"| R22
R22 ---|"parent (0,N)"| E10

R23{"R23: MEMBER_OF<br/>(M:N)"}:::rel
E3 ---|"(0,N)"| R23
R23 ---|"(1,N)"| E9
R23 --- R23_A1(["Role (S)"]):::attrS
R23 --- R23_A2(["Start_Date (S)"]):::attrS
R23 --- R23_A3(["End_Date (S)"]):::attrS
```